

Numerical Example of K-means Clustering:

As shown in Table 1, a small dataset contains the marks of 6 students, S1 to S6, for two subjects, namely, Math and Science. Our task is to cluster these students based on their marks into different groups using the K-means clustering algorithm.

a

Table 1: The small dataset for K-means Clustering

Student No.	Math Mark	Science Marks
S1	40	45
S2	60	75
S3	65	70
S4	90	90
S5	100	95
S6	45	35

In the first iteration, randomly take marks of students S6, S2, and S3 as the centroids of the initial 3 clusters (i.e., $k=3$). Denote the 3 centroids as C1 (45, 35), C2 (60, 75), and C3 (65, 70), respectively. We will then calculate the Euclidean distance from each of the samples to the centroids.

Calculate the distance from C1 to all the 6 samples:

$$\text{Distance from C1 to S1} = \sqrt{(45 - 40)^2 + (35 - 45)^2} = 11.18$$

$$\text{Distance from C1 to S2} = \sqrt{(45 - 60)^2 + (35 - 75)^2} = 42.72$$

$$\text{Distance from C1 to S3} = \sqrt{(45 - 65)^2 + (35 - 70)^2} = 40.31$$

$$\text{Distance from C1 to S4} = \sqrt{(45 - 90)^2 + (35 - 90)^2} = 71.06$$

$$\text{Distance from C1 to S5} = \sqrt{(45 - 100)^2 + (35 - 95)^2} = 81.39$$

$$\text{Distance from C1 to S6} = \sqrt{(45 - 45)^2 + (35 - 35)^2} = 0$$

Calculate the distance from C2 to all the 6 samples:

$$\text{Distance from C2 to S1} = \sqrt{(60 - 40)^2 + (75 - 45)^2} = 36.06$$

$$\text{Distance from C2 to S2} = \sqrt{(60 - 60)^2 + (75 - 75)^2} = 0$$

$$\text{Distance from C2 to S3} = \sqrt{(60 - 65)^2 + (75 - 70)^2} = 7.07$$

$$\text{Distance from C2 to S4} = \sqrt{(60 - 90)^2 + (75 - 90)^2} = 33.54$$

$$\text{Distance from C2 to S5} = \sqrt{(60 - 100)^2 + (75 - 95)^2} = 44.72$$

$$\text{Distance from C2 to S6} = \sqrt{(60 - 45)^2 + (75 - 35)^2} = 42.72$$

Calculate the distance from C3 to all the 6 samples:

$$\text{Distance from C3 to S1} = \sqrt{(65 - 40)^2 + (70 - 45)^2} = 35.36$$

$$\text{Distance from C3 to S2} = \sqrt{(65 - 60)^2 + (70 - 75)^2} = 7.07$$

$$\text{Distance from C3 to S3} = \sqrt{(65 - 65)^2 + (70 - 70)^2} = 0$$

$$\text{Distance from C3 to S4} = \sqrt{(65 - 90)^2 + (70 - 90)^2} = 32.02$$

$$\text{Distance from C3 to S5} = \sqrt{(65 - 100)^2 + (70 - 95)^2} = 43.01$$

$$\text{Distance from C3 to S6} = \sqrt{(65 - 45)^2 + (70 - 35)^2} = 40.31$$

All the calculated Euclidean distances in the first iteration of K-means algorithm are presented in Table 2.

Table 2: The calculated Euclidean distances from each sample to the 3 centroids in the first iteration of K-means clustering

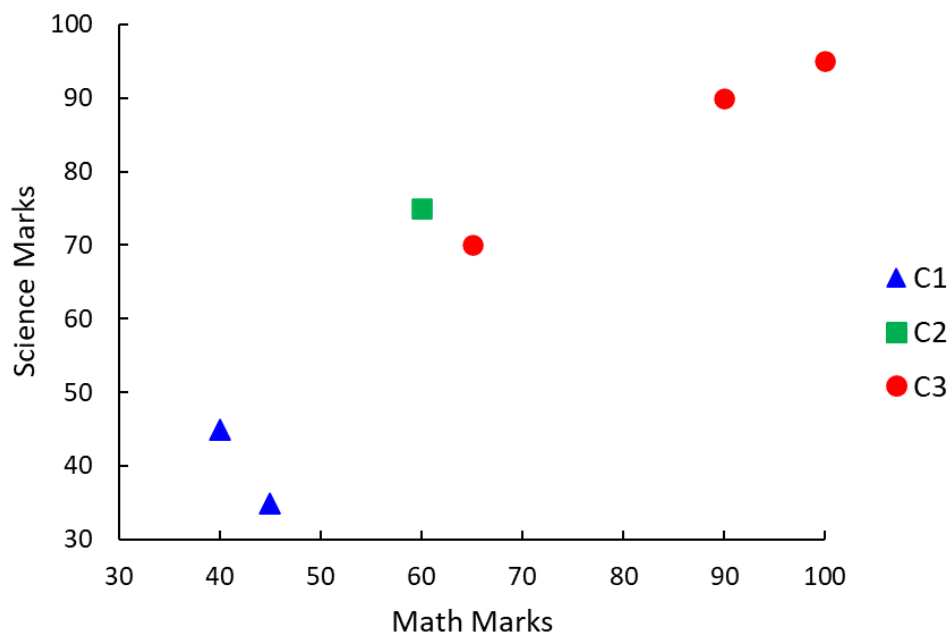
	C1	C2	C3
S1	11.18	36.06	35.36
S2	42.72	0	7.07
S3	40.31	7.07	0
S4	71.06	33.54	32.02
S5	81.39	44.72	43.01
S6	0	42.72	40.31

Using the data in Table 2, we can determine which sample is assigned to which cluster in the first iteration of the K-means algorithm based on its distance to the 3 centroids. As shown in Table 3, the samples are assigned to the closest centroids. For instance, S1 is closer to C1 than to C2 and C3, so it is assigned to C1; S2 is closer to C2 than to C1 and C3, so it is assigned to C2; S3 is closer to C3 than to C1 and C2, so it is assigned to C3.

Table 3: The assigned centroid for each sample in the first iteration of K-means clustering

	C1	C2	C3	Assigned Centroid
S1	11.18	36.06	35.36	C1
S2	42.72	0	7.07	C2
S3	40.31	7.07	0	C3
S4	71.06	33.54	32.02	C3
S5	81.39	44.72	43.01	C3
S6	0	42.72	40.31	C1

The visual representation for the clustering as of now is as follows:



After clustering the samples, we need to recalculate the new centroid for each cluster by taking mean of samples assigned to that cluster.

New centroid C1 is the mean of S1 and S6:

$$C1 = \left(\frac{40+100}{2}, \frac{45+95}{2} \right) = (70, 70)$$

New centroid C2 is the mean of S2:

$$C2 = \left(\frac{60}{1}, \frac{75}{1} \right) = (60, 75)$$

New centroid C3 is the mean of S3, S4, and S5:

$$C3 = \left(\frac{65+90+100}{3}, \frac{70+90+95}{3} \right) = (85, 85)$$

Now, we can proceed with the second iteration of K-means clustering algorithm with the newly updated C1, C2, and C3.

Calculate the distance from C1 to all the 6 samples:

$$\text{Distance from C1 to S1} = \sqrt{(42.5 - 40)^2 + (40 - 45)^2} = 5.59$$

$$\text{Distance from C1 to S2} = \sqrt{(42.5 - 60)^2 + (40 - 75)^2} = 39.13$$

$$\text{Distance from C1 to S3} = \sqrt{(42.5 - 65)^2 + (40 - 70)^2} = 37.5$$

$$\text{Distance from C1 to S4} = \sqrt{(42.5 - 90)^2 + (40 - 90)^2} = 68.97$$

$$\text{Distance from C1 to S5} = \sqrt{(42.5 - 100)^2 + (40 - 95)^2} = 79.57$$

$$\text{Distance from C1 to S6} = \sqrt{(42.5 - 45)^2 + (40 - 35)^2} = 5.59$$

Calculate the distance from C2 to all the 6 samples:

$$\text{Distance from C2 to S1} = \sqrt{(60 - 40)^2 + (75 - 45)^2} = 36.06$$

$$\text{Distance from C2 to S2} = \sqrt{(60 - 60)^2 + (75 - 75)^2} = 0$$

$$\text{Distance from C2 to S3} = \sqrt{(60 - 65)^2 + (75 - 70)^2} = 7.07$$

$$\text{Distance from C2 to S4} = \sqrt{(60 - 90)^2 + (75 - 90)^2} = 33.54$$

$$\text{Distance from C2 to S5} = \sqrt{(60 - 100)^2 + (75 - 95)^2} = 44.72$$

$$\text{Distance from C2 to S6} = \sqrt{(60 - 45)^2 + (75 - 35)^2} = 42.72$$

Calculate the distance from C3 to all the 6 samples:

$$\text{Distance from C3 to S1} = \sqrt{(85 - 40)^2 + (85 - 45)^2} = 60.21$$

$$\text{Distance from C3 to S2} = \sqrt{(85 - 60)^2 + (85 - 75)^2} = 26.93$$

$$\text{Distance from C3 to S3} = \sqrt{(85 - 65)^2 + (85 - 70)^2} = 25.0$$

$$\text{Distance from C3 to S4} = \sqrt{(85 - 90)^2 + (85 - 90)^2} = 7.07$$

$$\text{Distance from C3 to S5} = \sqrt{(85 - 100)^2 + (85 - 95)^2} = 18.03$$

$$\text{Distance from C3 to S6} = \sqrt{(85 - 45)^2 + (85 - 35)^2} = 64.03$$

All the calculated Euclidean distances in the second iteration of K-means algorithm are presented in Table 4.

Table 4: The calculated Euclidean distances from each sample to the 3 centroids in the second iteration of K-means clustering

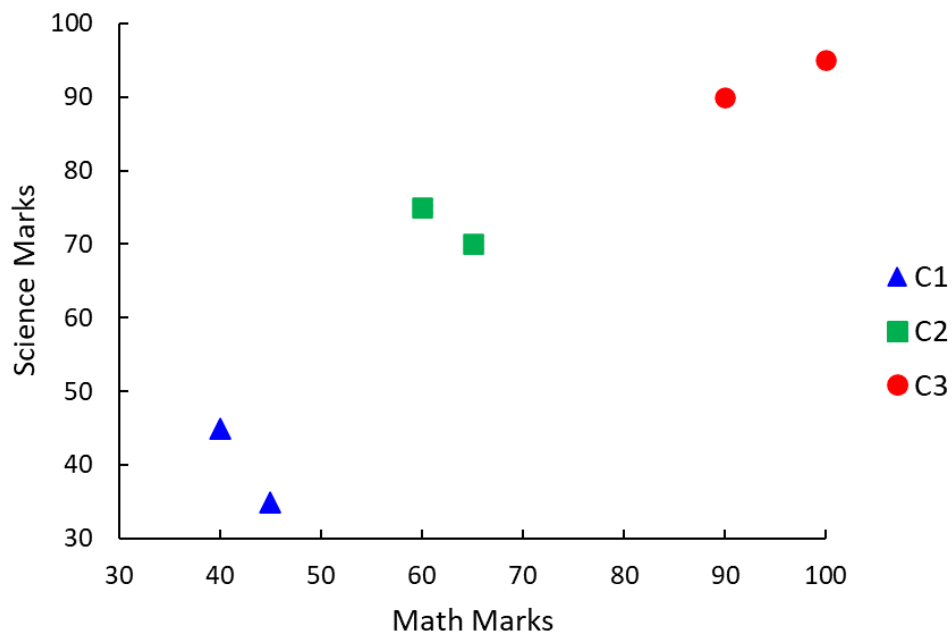
	C1	C2	C3
S1	5.59	36.06	60.21
S2	39.13	0	26.93
S3	37.5	7.07	25
S4	68.97	33.54	7.07
S5	79.57	44.72	18.03
S6	5.59	42.72	64.03

Using the data in Table 4, we can update which sample is assigned to which cluster in the second iteration of the K-means algorithm based on its distance to the 3 centroids. For instance, as shown in Table 5, S3 is closest to C2, and it is now assigned to C2 (note that S3 was assigned to C3 in the first iteration).

Table 5: The assigned centroid for each sample in the second iteration of K-means clustering

	C1	C2	C3	Assigned Centroid
S1	5.59	36.06	60.21	C1
S2	39.13	0	26.93	C2
S3	37.5	7.07	25	C2
S4	68.97	33.54	7.07	C3
S5	79.57	44.72	18.03	C3
S6	5.59	42.72	64.03	C1

The visual representation for the clustering as of now is as follows:



After clustering the samples, we need to recalculate the new centroid for each cluster by taking mean of samples assigned to that cluster.

New centroid C1 is the mean of S1 and S6:

$$C1 = \left(\frac{40+45}{2}, \frac{45+35}{2} \right) = (42.5, 40)$$

New centroid C2 is the mean of S2 and S3:

$$C2 = \left(\frac{60+65}{2}, \frac{75+70}{2} \right) = (62.5, 72.5)$$

New centroid C3 is the mean of S4 and S5:

$$C3 = \left(\frac{90+100}{2}, \frac{90+95}{2} \right) = (95, 92.5)$$

Now, we can proceed with the third iteration of K-means clustering algorithm with the newly updated C1, C2, and C3.

Calculate the distance from C1 to all the 6 samples:

$$\text{Distance from C1 to S1} = \sqrt{(42.5 - 40)^2 + (40 - 45)^2} = 5.59$$

$$\text{Distance from C1 to S2} = \sqrt{(42.5 - 60)^2 + (40 - 75)^2} = 39.13$$

$$\text{Distance from C1 to S3} = \sqrt{(42.5 - 65)^2 + (40 - 70)^2} = 37.5$$

$$\text{Distance from C1 to S4} = \sqrt{(42.5 - 90)^2 + (40 - 90)^2} = 68.97$$

$$\text{Distance from C1 to S5} = \sqrt{(42.5 - 100)^2 + (40 - 95)^2} = 79.57$$

$$\text{Distance from C1 to S6} = \sqrt{(42.5 - 45)^2 + (40 - 35)^2} = 5.59$$

Calculate the distance from C2 to all the 6 samples:

$$\text{Distance from C2 to S1} = \sqrt{(62.5 - 40)^2 + (72.5 - 45)^2} = 35.53$$

$$\text{Distance from C2 to S2} = \sqrt{(62.5 - 60)^2 + (72.5 - 75)^2} = 3.54$$

$$\text{Distance from C2 to S3} = \sqrt{(62.5 - 65)^2 + (72.5 - 70)^2} = 3.54$$

$$\text{Distance from C2 to S4} = \sqrt{(62.5 - 90)^2 + (72.5 - 90)^2} = 32.6$$

$$\text{Distance from C2 to S5} = \sqrt{(62.5 - 100)^2 + (72.5 - 95)^2} = 43.73$$

$$\text{Distance from C2 to S6} = \sqrt{(62.5 - 45)^2 + (72.5 - 35)^2} = 41.38$$

Calculate the distance from C3 to all the 6 samples:

$$\text{Distance from C3 to S1} = \sqrt{(95 - 40)^2 + (92.5 - 45)^2} = 72.67$$

$$\text{Distance from C3 to S2} = \sqrt{(95 - 60)^2 + (92.5 - 75)^2} = 39.13$$

$$\text{Distance from C3 to S3} = \sqrt{(95 - 65)^2 + (92.5 - 70)^2} = 37.5$$

$$\text{Distance from C3 to S4} = \sqrt{(95 - 90)^2 + (92.5 - 90)^2} = 5.59$$

$$\text{Distance from C3 to S5} = \sqrt{(95 - 100)^2 + (92.5 - 95)^2} = 5.59$$

$$\text{Distance from C3 to S6} = \sqrt{(95 - 45)^2 + (92.5 - 35)^2} = 76.2$$

All the calculated Euclidean distances in the third iteration of K-means algorithm are presented in Table 6.

Table 6: The calculated Euclidean distances from each sample to the 3 centroids in the third iteration of K-means clustering

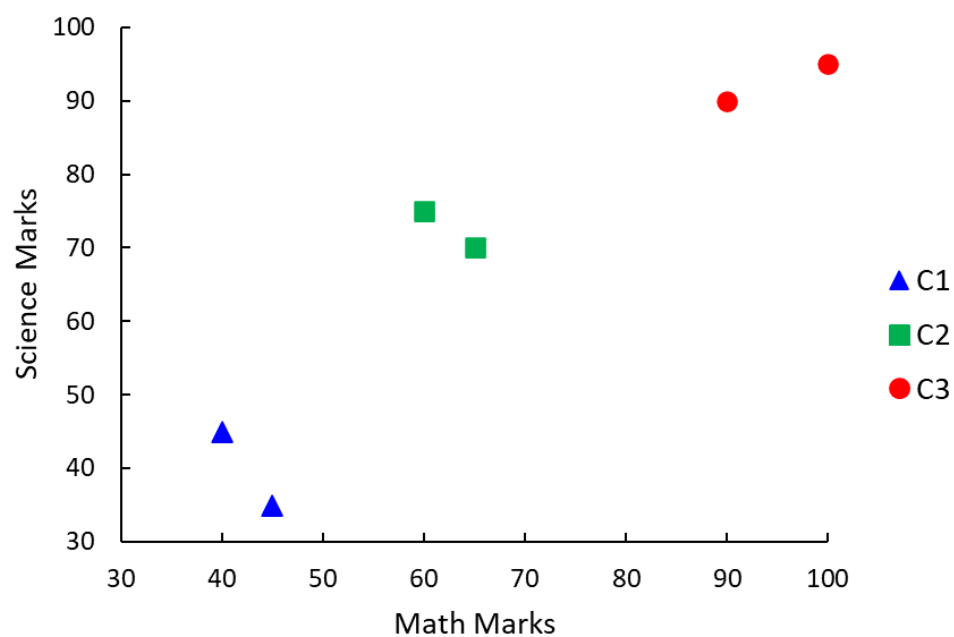
	C1	C2	C3
S1	5.59	35.53	72.67
S2	39.13	3.54	39.13
S3	37.5	3.54	37.5
S4	68.97	32.6	5.59
S5	79.57	43.73	5.59
S6	5.59	41.38	76.2

Using the data in Table 6, we can update which sample is assigned to which cluster in the second iteration of the K-means algorithm based on its distance to the 3 centroids.

Table 7: The assigned centroid for each sample in the third iteration of K-means clustering

	C1	C2	C3	Assigned Centroid
S1	5.59	35.53	72.67	C1
S2	39.13	3.54	39.13	C2
S3	37.5	3.54	37.5	C2
S4	68.97	32.6	5.59	C3
S5	79.57	43.73	5.59	C3
S6	5.59	41.38	76.2	C1

The visual representation for the clustering as of now is as follows:



From Table 7 and the visual representation above, we can see that the cluster assignments for all samples remain unchanged in the third iteration compared to the second iteration. As a result, the K-means clustering algorithm converges and stops here. The final centroids are C1 (42.5, 40), C2 (62.5, 72.5), and C3 (95, 92.5).